AVIRAL NAUTIYAL

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Education

M.S. in Computer Science (3.9/4.0) — **Sept. 2024 – May 2026**
Lakehead University — *Thunder Bay, ON, Canada*
- **Focus areas:** Machine Learning, Deep Learning, NLP, Cloud Computing (AWS), Data Structures & Algorithms
- **Master's Research Project:** Brain Tumor Detection from MRI images using Machine Learning

Master of Computer Application (8.42/10) — **Aug. 2022 – Jul. 2024**
DIT University — *Dehradun, Uttarakhand, India*
- **Core Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks
- **Advanced Coursework:** Java Programming, Software Engineering, Artificial Intelligence, Machine Learning

Bachelor of Science (Physical Science with Computer Science) (8.9/10) — **Aug. 2019 – Jun. 2022**
University of Delhi — *New Delhi, India*
- **Core Coursework:** Database Management Systems, Operating Systems, Computer System Architecture, R Programming, HTML, Calculus and Matrices, Algebra, Data Structures, Computer Networks, Discrete Mathematics, Probability and Statistics

Technical Skills

Programming & Database: Python (proficient), Java, R, SQL
Development Tools & Cloud: Git, GitHub, GitLab, GitHub Actions (CI/CD), Docker, AWS, Firebase, Gradio, VS Code, Jupyter Notebooks, LaTeX
Machine Learning & AI: Deep Learning (CNNs, Transformers), Computer Vision, Transfer Learning, Feature Selection & Engineering, Metaheuristic Optimization, NLP, Model Evaluation, Data Preprocessing
ML & Data Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
Platforms: macOS, Windows, Linux (Ubuntu)
Methodologies & Soft Skills: Cross-functional Collaboration, Problem Solving, Communication, Agile/Scrum, Time Management, Adaptability

Projects

Brain Tumor Detection from MRI images using Machine Learning | | | *Python, TensorFlow, Scikit-learn*
- Implemented published **Black Hole Optimization (BHO) feature selection** for brain tumor MRI classification, reducing a **3,840-dimensional fused feature vector** to approximately **2,000 optimal features (∼48% reduction).**
- Fused deep features from **ResNet50, VGG16, and EfficientNet-B0** as the optimization input, improving computational efficiency while preserving classification performance on MRI images.
- Implemented and evaluated an **Optimized Stochastic Gradient Descent based Logistic Regression classifier** across **3 MRI datasets**, achieving up to **98.48% classification accuracy** with significantly **lower testing time** than traditional models. Deployed it as a live, containerized (using Docker) web demo on Google Cloud Run with GitHub Actions CI/CD pipelines.

Traffic Sign Recognition System | | *Python, TensorFlow, OpenCV*
- Developed a **CNN-based multi-class image classification system** to recognize **43 traffic sign classes** using the GTSRB dataset.
- Implemented image preprocessing pipelines on **50,000 labelled traffic sign images** including normalization, resizing, and data augmentation techniques.
- Designed and trained deep CNN architectures with regularization techniques and structured training/validation workflows **(early stopping, converging within 20-30 epochs)**, achieving **approximately 97% classification accuracy** on unseen test data.

AI-Based Facial Recognition Attendance System | | *Python, OpenCV, Firebase*
- Built a **real-time computer vision system** for facial recognition-based attendance tracking, with a pipeline for face detection, feature encoding, and similarity matching, validated on a dataset of **100 labelled face images**.
- Integrated the application with **Firebase cloud-based storage and databases** for persistent attendance records and student metadata.
- Achieved **approximately 80–85% recognition accuracy** under controlled lighting conditions during system evaluation.